Performance characteristics of the DPMs

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1 Abstract

Los DPM son entrenados para la determinación de los síntomas clínicos AD mediante la selección de medidas neuropsicológicas. Esta información junto con los biomarcadores de core 1 y 2 reforzará el diagnóstico AD.

Los DPM podrían servir para: a) Validar modelos teóricos, b) Análisis de factores de riesgo, c) Diagnóstico y pronóstico de los pacientes y d) Analizar cómo captura un marcador la progresión de la enfermedad

This work studies the robust modeling of Alzheimer's disease progression using parametric methods.

2 Introduction

Alzheimer's disease (AD) is the most common neurodegenerative disorder in elderly individuals. Neuropsychological assessments are employed to evaluate its impact on cognition and behavior. AD exhibits a non-linear progression spanning decades, posing a challenge for clinicians in predicting its course. Disease progression modeling (DPM) uses longitudinal studies to develop data-driven quantitative models that describe the evolution of the disease over time. DPMs applied to AD were inspired by theoretical models proposing the evolution of biomarkers across the different stages of the AD [1, 2].

One of the objectives is to validate these models through longitudinal patient studies in both the preclinical and prodromal phases.

// También permite para el análisis de factores de riesgo // Asistencia en el diagnóstico y pronóstico de los pacientes

Additionally, DPMs are also applied in the automation of diagnosis, allowing estimation of the remaining time for a patient to reach a specific clinical event. For instance, in the preclinical phase, it is relevant to calculate the estimated time remaining for an individual with a normal cognitive state to initiate cognitive decline. So, it is crucial to develop robust DPMs for a more comprehensive understanding of the disease, leading to a significant improvement in patient diagnosis and monitoring.

The construction of DPMs faces the double problem that not all measurements of the markers are available in every visit (i.e. handling missing values), and that the time intervals between patient visits tend to be irregular. Several approaches have been employed in constructing DPMs, ranging from parametric models [3, 4], which assumed that the markers' temporal trajectories are modeled by quasi-sigmoidal functions, to frameworks based on deep learning [5]. The former are simpler and require less data for DPM construction compared to the latter. However, non-parametric models do not make assumptions about the type of function for trajectories, although interpreting their results is more challenging and training them with sparse and irregular temporal data is difficult.

Depending on the markers used to train a DPM, the proposed natural history by such an ap-

proach could vary. It is suggested to choose a set of features that allows the constructed DPM to closely resemble the actual evolution of the disease. For this purpose, various measures related to patient diagnosis and prognosis will be considered. The most suitable collection of patient measures over time will be selected, ensuring that the DPM trained with these markers provides the most accurate estimation of disease progression.

We use here a longitudinal data sets of patients who were diagnosed with Alzheimer's disease at some point of the study to reconstruct the natural history of the disease. We construct an AD Course Map showing the reduction in glucose consumption across brain regions, the thinning of the cerebral cortex, the deformation of the hippocampus due to atrophy, and the progressive onset of cognitive deficit from the preclinical to the clinical phase of the disease... This statistical analysis aims to highlight how genetic, biological or environmental factors change disease progression profiles. Once the map is built, we position new individuals into the map and then read how data will change in the coming years.

3 Methods

3.1 Parametric DPM

We denote the biomarker measurement k for individual i at time j as y_{ijk}, where $i = 1, \ldots, n$, $k = 1, \ldots, p$ and $j = 1, \ldots, q_{ik}$. The model can be written as:

$$y_{ijk} = f(s_{i,j}, \Theta_k) + e_{ijk},$$

where f corresponds to the model, $z_{i,j}$ are latent random variables associated to the i-th subject, Θ_k are the model parameters, which include variables from f and $s_{i,j}$, and e_{ijk} is an error term.

In general, the aim is to identify the parameters that most accurately represent the observations, such as those that optimize the likelihood [] or optimization the robust nonlinear regression [].

A medida que se tenga más muestras del sujeto, más particular serán las trayectorias del paciente. Por el contrario, con pocas visitas, las trayectorias de sus marcadores tienden según el modelo de la población en general.

3.2 RPDPM

The latent variable $s_{i,j}$ transforms the patient's age $t_{i,j}$ into a timescale that defines the progression of the disease, where α_i and β_i represent the temporal parameters for the i-th patient:

$$s_{i,j} = \alpha_i \cdot t_{i,j} + \beta_i.$$

The geometric model for the trajectories is defined by:

$$f(s; \Theta) = (a - d)g(s; \Theta) + d$$

where a and d are normalization parameters for the markers, and $g(s; \Theta)$ represents a logistic function. Finally, the multiobjective optimization for robust nonlinear regression is defined as:

$$\left\{\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\Theta}}\right\} = \min_{i,j,k} \sum_{i,j,k} \omega_i \rho \left(\frac{y_{ijk} - f(s_{i,j}, \Theta_k)}{\sigma_k} \right)$$

where $\rho(\cdot)$ is a maximum likelihood-type function and $\omega_i = 1/N_i$ is a weighting factor for normalizing the objective function with the number of available points per subject (N_i).

3.3 Leaspy

Disease course mapping is a statistical technique which allows the decomposition of these two types of variations seen in a longitudinal data set. It builds on abstract geometric principles so that it is not specific to particular data11,12. It combines the concept of time-warp with translations of curves.

The atlas takes the form a series of progression curves for each biomarker where the time axis represents a disease stage, which we call here an Alzheimer Age (AA). Subject's data take the form the biomarkers measurements at several timepoints. The transformation of model to the

subject's data consist in computing (1) a time-shift, which translates the model along the time axis to accommodate for changes in age at disease onset, (2) an acceleration factor,which scales the time interval to accommodate for differences in speed of progression, and (3) intermarker spacings, which translate each biomarker differently to accommodate for differences in the timing and ordering among biomarkers. The first two parameters define a time-warp function that maps the actual age of the subject to his AA. This function therefore changes the dynamics of progression but not the trajectory. In turn, inter-marker spacings shift the patient trajectory to account for phenotypic differences across subjects, as illustrated in Figure 1.

Time-shifts and acceleration factor plays the same role of adjusting for the dynamics of change. They map the actual age of the subject at each visit to its Alzheimer Age that is a time-point in the reference scenario. Intermarker spacing adjust for the spatiotemporal.

3.4 Grid Search

Dado que cada combinación de marcadores que son utilizados para entrenar un DPM propone una historia natural, se propone utilizar un metodo GridSearchCV de forma que las medidas utilizadas para evaluar los DPM localice la menjor construcción del DPM.

We applied GridSearchCV which uses the Grid Search technique for finding the optimal parameters to increase the model performance. Grid Search uses a different combination of all the specified hyperparameters and their values and calculates the performance for each combination and selects the best value for the hyperparameters. This makes the processing time-consuming and expensive based on the number of hyperparameters involved.

4 DPM's performances

Assessing the performance of DPM is essential. This evaluation can be carried out through two types of measures: a) Measuring the accuracy in estimating the temporal trajectories of patients' markers. b) Evaluating patients' diagnosis over time, including estimates of remaining time for key clinical events, such as the age at which cognitive decline begins.

4.1 Evaluations of estimated trajectories

The Mean Absolute Error (MAE) is a performance measure for models that is less sensitive to outliers (Chai and Draxler, 2014), calculated based on the absolute differences between actual and estimated values:

$$MAE_k = \frac{1}{N_k} \sum \left| y_{y,j,k} - \hat{y}_(i,j,k) \right|$$

N_k represents the total number of measurements for the k-th marker across all subjects and visits. The measure and estimated value of this marker for the i-th subject at the j-th visit are denoted by $y_{i,j,k}$ and $\hat{y}(i,j,k)$ respectively. To enable cross-variable comparisons, the MAEs are normalized as NMAE [93], as in Equation (13)

4.2 Clinical status classification

$s_{i,j}$ is the disease progression score (DPS) for the j-th visit of the i-th subject and is obtained as:

$$s_{i,j} = \alpha_i \cdot t_{i,j} + \beta_i,$$

where $t_{i,j}$ is the age of subject i in visit j, and α_i and β_i are the rate and onset of disease progression of subject i, respectively.

The clinical status is classified based on the DPSs with a Bayesian classifier that uses the kernel density estimation (KDE) (Parzen, 1962)-based fitted likelihoods as:

$$p(s|c) = \frac{1}{n \cdot h} \sum_{t=1}^{n} K\left(\frac{s - s_t}{h}\right)$$

where $K(\cdot)$ is the kernel, $h \geq 0$ is a smoothing parameter and c denotes a clinical group. The predicted class for observation $s_{i,j}$ is c_r, which maximizes the posterior probability:

$$r = \arg\max_r \left(p(c_r) \cdot p(s_{i,j}|c_r) \right).$$

4.3 Measures regarding the timeframes of conversion toward cognitive decline

Quality measures were proposed for DPMs associated with a specific clinical event: in this case, the onset of cognitive decline in a patient. Firstly, the estimated age at which the subject initiates this decline was calculated according to a proposed model. This date was determined as the average time between the first visit diagnosed with MCI and the preceding visit where the patient was considered to have normal cognitive function. Additionally, the patient's final visit in the

longitudinal study must confirm the patient's cognitive decline or dementia status. Based on the proposed date for this clinical event in the i-patient, denoted as $Age_{i_{onset}}$, the following measures are suggested.

For sCU subjects, the age at the event must be greater than the patient's age at the last visit:

$$SCORE_1 = \frac{\#\left\{i|(i \in sCU) \cap ((Age_{i_{end}} < Age_{i_{onset}})\right\}}{\#sCU}$$

For sMCI subjects, the age at event must be the same as the age at baseline:

$$SCORE_2 = \frac{\#\left\{i|(i \in sMCI) \cap ((Age_{i_1} < Age_{i_{onset}})\right\}}{\#sMCI}$$

Regarding subjects progressing from CU to MCI, we measured: a) the percentage of preclinical CU cases detected by the DPMs:

$$SCORE_3 = \frac{\#\left\{i|(i \in pCU) \cap ((Age_{i_1} < Age_{i_{onset}})\right\}}{\#pCU}$$

$$SCORE_4 = \frac{\#\left\{i|(i \in pCU) \cap ((Age_{i_{end}} \geq Age_{i_{onset}})\right\}}{\#pCU}$$

b) the correlation between age at the event and its estimation by the model, and c) the correlation between the years remaining for the individual to reach cognitive decline according to clinical diagnosis and the model's estimation from the visit's date at baseline.

4.4 Data partitions and boostraping

To assess the algorithms, the dataset is divided into two non-overlapping sets for training and testing. Furthermore, bootstrapping is employed in the experiments to evaluate the DPM's performances. Within each bootstrap iteration, a subset of the training subjects is randomly sampled with replacement, taking into account their clinical grouping, which ensures that each bootstrap sample includes data representing all diagnostic statuses.

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